Cambridge (CIE) A Level Maths: Probability & Statistics 1

Your notes

Working with Data

Contents

- Interpreting Data
- Skewness

Your notes

Interpreting Data

You may be asked to comment on how statistics could affect data or how removing or adding a new piece of data could change statistics you have calculated. You may also be asked to compare two data sets of a similar context.

Analysing statistics calculated from a data set:

- You could be asked to use some statistics you have calculated, such as the median or interquartile range, to make decisions about the set of data you were analysing
  - Writing about the mean or median gives information about where the data is located
  - Writing about the range, interquartile range, standard deviation or variance gives information about how spread out the data is
    - A lower value means the data set is more consistent
    - A greater value means the data set is more spread out, or varied
- When commenting on a data set you should usually write about both a measure of location and a measure of spread
- You should pair the median with the range or interquartile range and the mean with the standard deviation or variance
  - Use the mean and standard deviation/variance when the data is roughly symmetrical and does not contain outliers
  - Use the median and interquartile range when the data contains outliers
- You should always write your analysis in the context of the data
  - Sometimes a lower mean/median is better: time taken to complete a puzzle
  - Sometimes a higher mean/median is better: score on a test

Changing a data set:

- Sometimes it might be discovered that a value was omitted from a data set and needs to be added in
- Similarly a data value could be found to be an error and will need to be removed (**cleaned**) from the data set
- It is important to be aware of how the **measures of location** and **spread** may change when the data is added or removed
  - Adding or removing a data value to the set could change the mean, median and quartiles
  - How each statistics changes will depend on where the data value lies within the data set

© 2025 Save My Exams, Ltd.

Your notes

    - Adding a data point below the mean, or removing one from above will cause the mean to decrease
    - Adding a data point above the mean, or removing one from below will cause the mean to increase
    - The median and quartiles may or may not change depending on the data value, you should always check these cases individually
- Adding or removing extreme values will change the value of the mean by a lot but affect the median in the same way as any other value

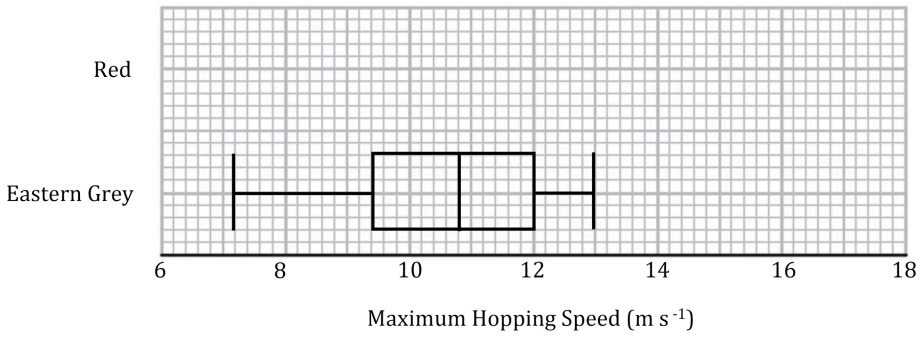
Comparing two data sets

- When comparing two data sets you must comment on both:
    - A **measure of location** such as the mean, median or mode and
    - A **measure of spread** such as the range, interquartile range, variance or standard deviation
        - If you comment on the mean as the measure of location you should use the standard deviation or variance as the measure of spread
        - If you comment on the median as the measure of location you should use the interquartile range as the measure of spread
- You should use information about the data to decide which measure of location and measure of spread is the best to use
    - If data contains extreme values then it is best to use the median and interquartile range to compare the data sets
    - Extreme values can cause the mean to be an unreliable statistic
- It is common to be asked to compare two data sets that are represented as two **box plots** on the same scale
    - You should write about both the median and the interquartile range
- When writing a comparison about the data you should always write in **context**

Worked Example

The diagram below shows a box plot of the average hopping speeds, in m s^{-1}, of some Eastern Grey kangaroos living in an Australian nature reserve.

Your notes

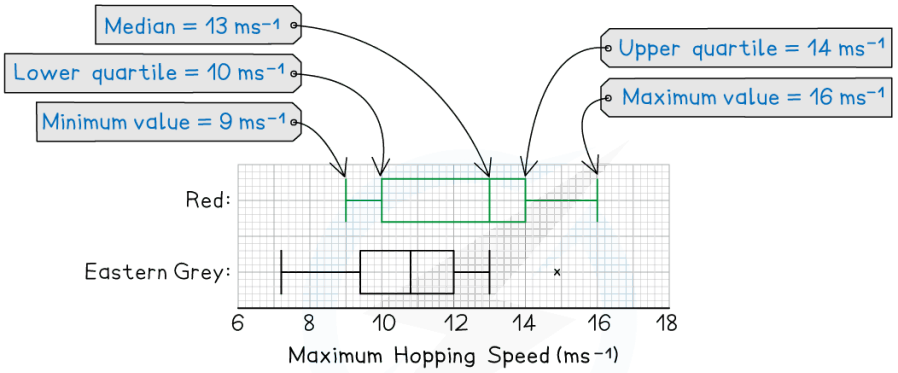


The maximum hopping speeds, in m s^{-1}, of some Red Kangaroos living in the nature reserve are summarised below.

Lower quartile: 10 Median: 13 Upper quartile: 14

Minimum value: 9 Maximum value: 16

Draw a box plot on the grid to represent the data for the Red Kangaroos and compare the distribution of the hopping speeds for the Red and Eastern Grey kangaroos.



To compare the distributions, comment on the median (to compare location) and the interquartile range (to compare spread).

The median hopping speed for the Red Kangaroos is 2.2 ms^{-1} faster than median hopping speed for the Eastern Grey Kangaroos. The interquartile range for the Red Kangaroos is 1.4 ms^{-1} greater than for the Eastern Grey Kangaroos. This suggests the Red Kangaroos are faster but with more varied hopping speeds than the Eastern Greys.

Copyright © Save My Exams. All Rights Reserved

Examiner Tips and Tricks

- Remember to always write about data sets and their distributions in the context of the question. The number of marks available in comparison question are often an indication of how much you should say in the answer.

Your notes

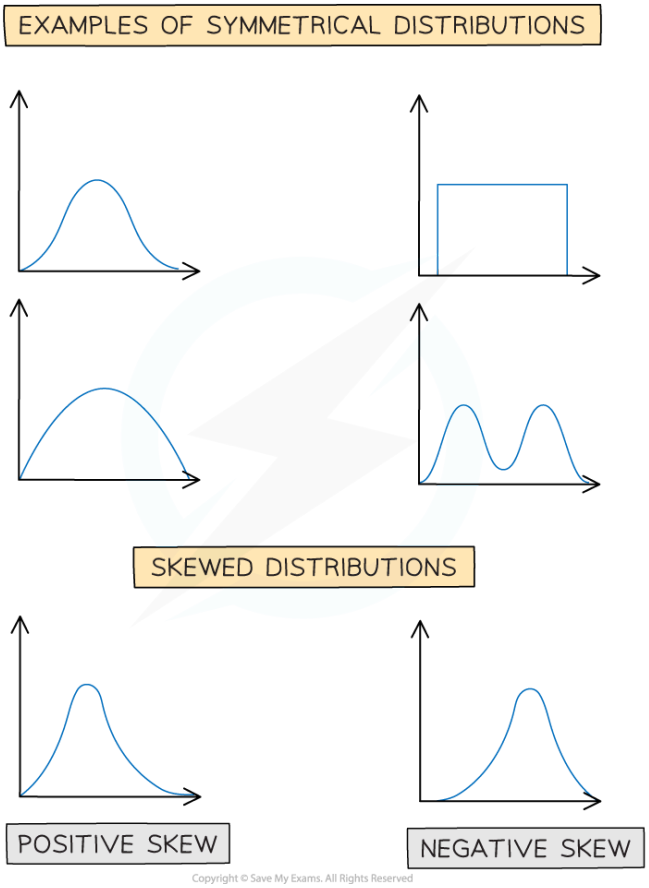
© 2025 Save My Exams, Ltd.

Your notes

Skewness

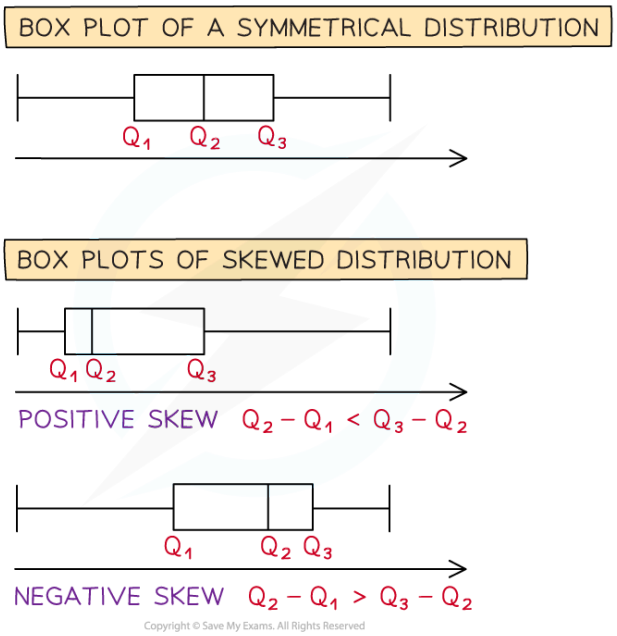
What is skewness?

- The distribution of a data set is either symmetrical or it has skewness.
- Skewness describes the way in which data in a non – symmetrical distribution is leaning
- A distribution that has its tail on the right side has **positive skew**
- A distribution that has its tail on the left side has **negative skew**

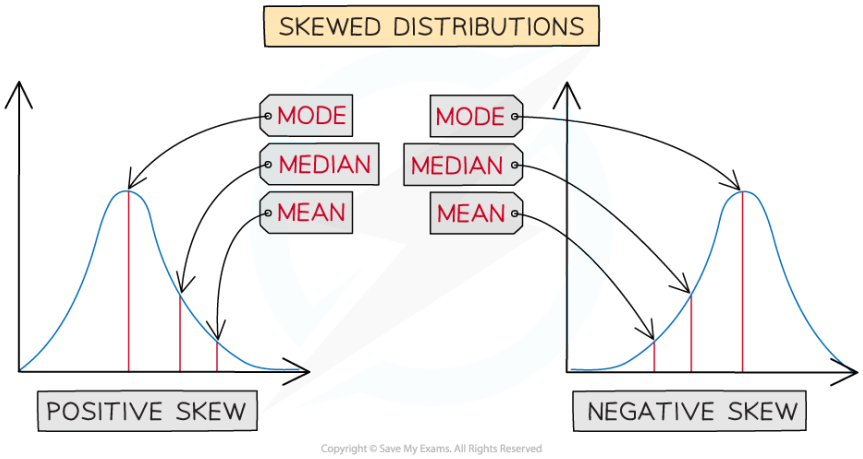


- If the distribution is shown on a box plot looking at the difference between the quartiles can help decide how it is skewed
  - If the median is closer to the lower quartile then the distribution has **positive skew**
    - $Q_3 - Q_2 > Q_2 - Q_1$
  - If the median is closer to the upper quartile then the distribution has **negative skew**
    - $Q_3 - Q_2 < Q_2 - Q_1$

Your notes



- Looking at the values of the statistics can help you decide whether distribution is positively skewed or negatively skewed
  - In a positively skewed distribution
    - Mode < median < mean
  - In a negatively skewed distribution
    - Mean < median < mode



Examiner Tips and Tricks

- You only need to be able to recognise the different types of skewness

- It can help to comment on skewness when asked to compare distributions

Your notes

© 2025 Save My Exams, Ltd.